

KnowSelf: Teaching LLM Agents When to Think, Reflect, or Seek Knowledge

Agentic knowledgeable self-awareness: a framework that lets LLM agents adaptively choose fast execution, reflective correction, or external knowledge retrieval based on task situation.

Current Agent Training Uses “Flood Irrigation”

Problem in Existing Paradigms

- Trajectory imitation injects full demonstrations regardless of need.
- Trial-and-error approaches add reflection everywhere, not selectively.
- Knowledge-augmented methods query knowledge in all steps (Know%=100%).
- Result: unnecessary context, higher compute, and potential action noise.

Why This Fails

- Different states require different cognition budgets.
- Humans switch between intuitive, deliberative, and knowledge-seeking modes.
- Agents lacking metacognitive routing overuse expensive modules.
- Goal: situationally decide **when not** to retrieve knowledge.

Three Situational Modes: Decision Tree for Agent Cognition

Fast Thinking

- Criterion: policy predicts correct next action directly.
- Behavior: execute immediately with minimal deliberation.
- Use case: routine state transitions and clear affordances.



Slow Thinking

- Criterion: initial action uncertain/incorrect but recoverable by reflection.
- Behavior: invoke self-reflection and revise action plan.
- Use case: ambiguity, long-horizon dependencies, planning slips.

Knowledgeable Thinking

- Criterion: environment state requires external facts not in parametric memory.
- Behavior: trigger retrieval/tool use, then continue planning with grounded evidence.
- Use case: commonsense/task priors (e.g., heating prerequisite, object placement rules).

KnowSelf Pipeline: Data Construction → Two-Stage Training → Self-Aware Inference

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Heuristic Situation Judgment

- Self-explore trajectories on ALFWorld tasks.
- Label states as fast / slow / knowledgeable situations.
- Insert special cognitive-routing tokens into traces.

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Two-Stage Optimization

- Stage I: supervised fine-tuning on token-augmented traces.
- Stage II: RPO to reinforce decision quality and task success.
- No costly human dense annotations required.

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Self-Aware Inference

- Model emits situational token online at each step.
- Token gates direct action, reflection, or retrieval module.
- Achieves selective knowledge usage with higher final returns.

Qualitative Evidence: Reflection and Knowledge Triggers in Action

Task A: “put a hot mug in cabinet”

- KnowSelf detects planning mismatch and triggers slow-thinking reflection.
- Revised plan includes heating step before placement.
- Successfully completes sequence; baseline OpenAI-O1 fails this trajectory.

Signal: reflective correction of initially insufficient plan.

Task B: “put a saltshaker in drawer”

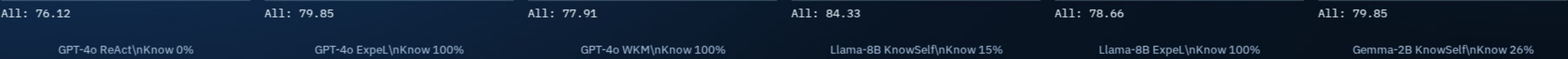
- KnowSelf identifies need for external procedural knowledge.
- Triggers knowledgeable-thinking path and retrieves task-relevant cue.
- Completes placement while DeepSeek-R1 becomes stuck.

Signal: retrieval only when endogenous policy is insufficient.

ALFWorld Results: KnowSelf Hits 84.33% with Only 15% Knowledge (Llama-8B)

Backbone	Method	Know%	pick&place	look_at_obj	pick_clean	pick_heat	pick_cool	pick_two	All
GPT-4o	ReAct	0	88.52	67.39	58.10	69.44	33.82	49.25	76.12
	Reflexion	0	89.34	64.34	58.57	66.22	30.88	51.49	73.73
	ExpeL	100	93.55	73.91	76.19	83.33	52.94	79.85	79.85
	ETO	0	86.89	64.78	60.48	69.28	35.29	50.75	74.48
	KnowAgent	100	90.98	65.65	67.86	77.03	51.47	68.66	77.61
	WKM	100	91.80	66.08	67.14	78.38	50.00	70.15	77.91
Llama-8B	KnowSelf	15	95.08	79.13	81.90	89.19	57.35	82.09	84.33
	ExpeL	100	92.62	72.17	74.29	81.08	51.47	77.61	78.66
Gemma-2B	KnowSelf	26	93.55	73.91	76.19	83.33	52.94	79.85	79.85
	ExpeL	100	90.16	66.08	69.52	77.03	47.06	72.39	76.87

Efficiency Advantage: 15–26% Knowledge Usage vs 100% Baselines



- KnowSelf attains highest accuracy while querying knowledge in only a minority of steps.
- Efficiency gain persists across model scales: Gemma-2B and Llama-8B both benefit.
- Empirical evidence supports selective retrieval over always-on retrieval.

Key Takeaways and Future Directions

- Indiscriminate knowledge injection is wasteful and can reduce downstream performance.
- Special-token supervision enables practical situational self-awareness without heavy annotation.
- KnowSelf (Llama-8B) outperforms strong GPT-4o baselines: 84.33 vs 76.12 (ReAct), 79.85 (Expel).
- Only 15–26% knowledge usage is needed, versus 100% for full-knowledge methods.
- Framework generalizes across model scales: Gemma-2B KnowSelf reaches 79.85.

Future Work

Extend self-aware routing to multi-agent coordination, continuous tool APIs, and adaptive confidence calibration under distribution shift.